

Troubleshooting SSH Connectivity and Splunk Forwarding Issues

Using Linux Logs, VirtualBox Networking, and Splunk Universal Forwarder

Project Type	Beginner SOC / Linux / Splunk Troubleshooting Project
Lab Environment	Ubuntu Server VM, Kali VM, Windows Host, Splunk Enterprise, Splunk Universal Forwarder
Main Log Source	/var/log/auth.log
Main Splunk Receiver	192.168.56.1:9997
Final Sourcetype	auth

Project Summary

This project documents the troubleshooting of SSH connectivity and Splunk log forwarding issues in a local cybersecurity lab. The investigation began when SSH was not working correctly from the client machine. The SSH service on Ubuntu was confirmed to be active, but the client initially returned a network error. After resolving the network connectivity issue, the Ubuntu server generated new SSH authentication logs locally, but Splunk did not immediately receive them. Further investigation showed that the Splunk Universal Forwarder was running but still configured with an old inactive receiver IP address. The forward-server was updated to the correct Windows Host-only IP address, and new SSH logs were successfully received in Splunk under the correct sourcetype.

1. Project Objective

The objective of this project was to troubleshoot SSH connectivity and Splunk log forwarding in a lab environment. The project focused on identifying whether the issue was caused by the SSH service, the virtual network configuration, the Linux authentication log, or the Splunk Universal Forwarder configuration.

Specific objectives included:

- Confirm whether the SSH service was installed, active, and listening on port 22.
- Troubleshoot the SSH connection error from Kali to Ubuntu.
- Review VirtualBox network adapter settings.
- Confirm that SSH authentication logs were being generated locally in `/var/log/auth.log`.
- Investigate why new authentication logs were not appearing in Splunk.
- Verify the Splunk Universal Forwarder status and forward-server configuration.
- Update the forward-server to the correct Splunk receiver IP address.
- Confirm that new logs were received in Splunk under sourcetype `auth`.

2. Lab Environment

Component	Description
Ubuntu Server VM	Target server running OpenSSH Server and generating <code>/var/log/auth.log</code> events.
Kali VM	Client machine used to initiate SSH connection attempts.
Windows Host	Host system running Splunk Enterprise and providing the VirtualBox Host-only network interface.
Splunk Universal Forwarder	Installed on Ubuntu to forward <code>/var/log/auth.log</code> events to Splunk Enterprise.
Splunk Enterprise	Used to search and analyze received SSH authentication logs.
VirtualBox Host-only Adapter	Used to allow communication between the host and virtual machines on the 192.168.56.x network.

3. Tools Used

- Ubuntu Terminal
- Kali Terminal
- Windows Command Prompt / PowerShell
- VirtualBox network settings
- Splunk Enterprise Search & Reporting
- Splunk Universal Forwarder
- Linux `auth.log`
- `nc` network connectivity test

4. Initial Problem

The initial problem was that SSH was not working as expected from the client machine. After investigating the SSH service on Ubuntu, the service was found to be active and listening on port 22. However, the client initially

returned a network-related error. Later, after new SSH logs were generated locally, the Splunk event counts did not increase, which suggested a separate forwarding issue.

5. Troubleshooting Methodology and Evidence

5.1 Checking SSH Service Status

The first step was to check whether the SSH service was running on the Ubuntu server.

sudo systemctl status ssh

```
kalafi@lifa-servers:~$ ip a
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d2:04:60 brd ff:ff:ff:ff:ff:ff
    inet 10.212.186.101/24 metric 100 brd 10.212.186.255 scope global dynamic enp0s3
        valid_lft 3011sec preferred_lft 3011sec
    inet6 fe80::a00:27ff:fed2:460/64 scope link
        valid_lft forever preferred_lft forever
kalafi@lifa-servers:~$ sudo tail -f /var/log/auth.log
2026-05-23T12:40:03.627017+00:00 lifa-servers sudo: pam_unix(sudo:session): session closed for user root
2026-05-23T12:41:05.906125+00:00 lifa-servers sudo: kalafi : TTY=ttty1 ; PWD=/home/kalafi ; USER=root ; COMMAND=/usr/bin/systemctl restart rsyslog
2026-05-23T12:41:05.906824+00:00 lifa-servers sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kalafi(uid=1000)
2026-05-23T12:41:06.001978+00:00 lifa-servers sudo: pam_unix(sudo:session): session closed for user root
2026-05-23T12:42:31.035464+00:00 lifa-servers sudo: kalafi : TTY=ttty1 ; PWD=/home/kalafi ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-05-23T12:42:31.035940+00:00 lifa-servers sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kalafi(uid=1000)
2026-05-23T12:45:01.065999+00:00 lifa-servers CRON[1205]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-05-23T12:45:01.077579+00:00 lifa-servers CRON[1205]: pam_unix(cron:session): session closed for user root
2026-05-23T12:45:58.452403+00:00 lifa-servers sudo: pam_unix(sudo:session): session closed for user root
2026-05-23T12:47:11.765871+00:00 lifa-servers sudo: kalafi : TTY=ttty1 ; PWD=/home/kalafi ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-05-23T12:47:11.774605+00:00 lifa-servers sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kalafi(uid=1000)
2026-05-23T12:55:01.149605+00:00 lifa-servers CRON[1226]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-05-23T12:55:01.172428+00:00 lifa-servers CRON[1226]: pam_unix(cron:session): session closed for user root
kalafi@lifa-servers:~$ ^C
kalafi@lifa-servers:~$ ^C
kalafi@lifa-servers:~$ sudo systemctl status ssh
ssh.service - OpenBSD Secure Shell server
    Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
    Active: active (running) since Sat 2026-05-23 12:36:23 UTC; 21min ago
    TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 858 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 866 (sshd)
    Tasks: 1 (limit: 4603)
    Memory: 2.2M (peak: 2.4M)
    CPU: 33ms
    CGroup: /system.slice/ssh.service
            └─866 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 23 12:36:23 lifa-servers systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 23 12:36:23 lifa-servers sshd[866]: Server listening on 0.0.0.0 port 22.
May 23 12:36:23 lifa-servers sshd[866]: Server listening on :: port 22.
May 23 12:36:23 lifa-servers systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Figure 1: SSH service status showing that OpenSSH server is active and listening on port 22.

The result showed that the SSH service was active and listening on port 22. This confirmed that the SSH service itself was not stopped.

5.2 SSH Connection Attempt from Kali

After confirming that SSH was running on Ubuntu, an SSH connection was attempted from Kali.

ssh kalafi@10.212.86.101



Figure 2: SSH connection attempt from Kali showing “Network is unreachable.”

The error indicated that the problem was related to network reachability or routing, not to SSH authentication.

5.3 Reviewing Virtual Machine Network Settings

The Ubuntu Server VM was shut down so that its VirtualBox network adapter settings could be reviewed safely.

```

OK ] Stopped ssh.service - OpenSSH Secure Shell server.
OK ] Stopped rsyslog.service - System Logging Service.
OK ] Stopped fwupd.service - Firmware update daemon.
OK ] Stopped session-1.scope - Session 1 of User kalafi.
OK ] Stopped systemd-random-seed.service - Load/Save OS Random Seed.
OK ] Removed slice system-getty.slice - Slice /system/getty.
OK ] Stopped target getty-pre.target - Preparation for Logins.
OK ] Stopped plymouth-quit-wait.service - Hold until boot process finishes up.
OK ] Stopping polkit.service - Authorization Manager...
OK ] Stopping systemd-logind.service - User Login Management...
OK ] Stopping user@1000.service - User Manager for UID 1000...
OK ] Stopped udisks2.service - Disk Manager.
OK ] Stopped polkit.service - Authorization Manager.
OK ] Stopped upower.service - Daemon for power management.
OK ] Stopped systemd-logind.service - User Login Management.
OK ] Stopped blk-availability.service - Availability of block devices.
OK ] Stopped user@1000.service - User Manager for UID 1000.
OK ] Stopped apport.service - automatic crash report generation.
OK ] Stopping systemd-user-sessions.service - Permit User Sessions...
OK ] Stopping user-runtime-dir@1000.service - User Runtime Directory /run/user/1000...
OK ] Unmounted run-user-1000.mount - /run/user/1000.
OK ] Stopped systemd-user-sessions.service - Permit User Sessions.
OK ] Stopped user-runtime-dir@1000.service - User Runtime Directory /run/user/1000.
OK ] Removed slice user-1000.slice - User Slice of UID 1000.
OK ] Stopped target remote-fs.target - Remote File Systems.
OK ] Stopped target remote-fs-pre.target - Preparation for Remote File Systems.
OK ] Stopped finalrd.service - Create final runtime dir for shutdown pivot root.
* Job SplunkForwarder.service/stop running (16s / 6min)

```

Figure 3: Ubuntu Server shutdown process before changing the virtual machine network settings.

The server was shut down before reviewing or changing the virtual network configuration.

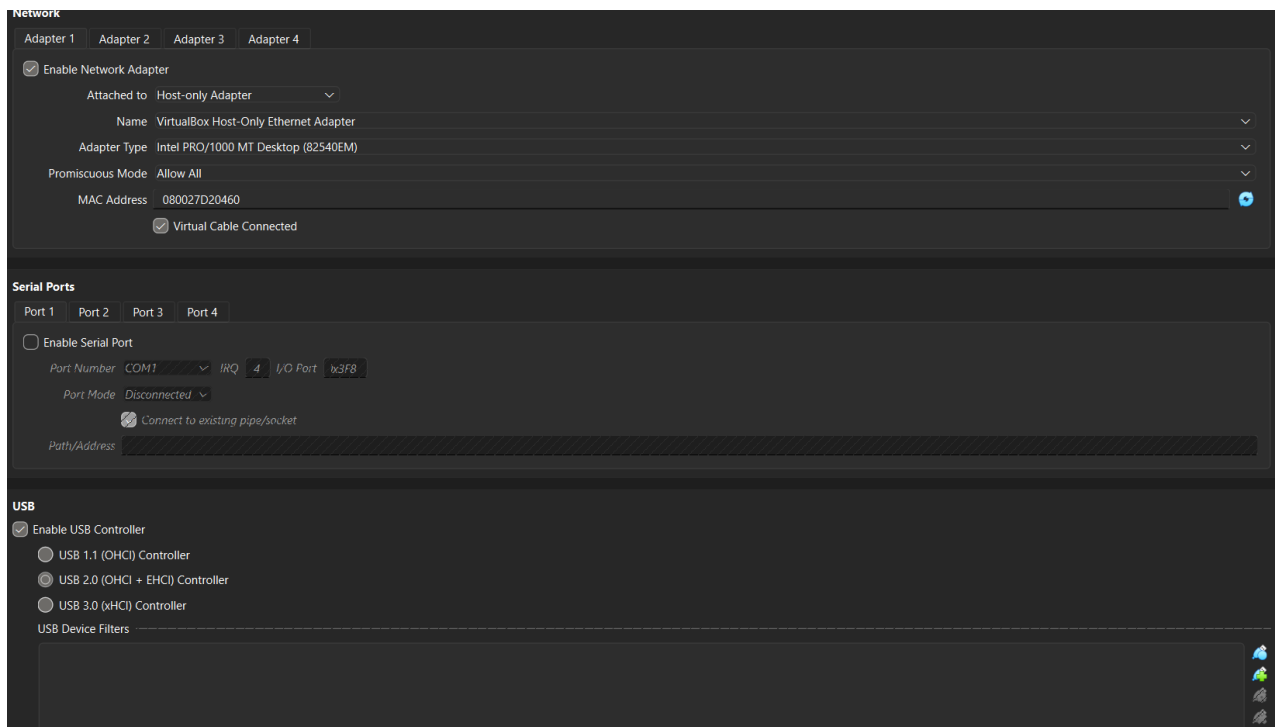


Figure 4: VirtualBox network adapter setting showing the VM attached to a Host-only Adapter.

The VM network setting showed that the adapter was attached to the VirtualBox Host-only Adapter. For VM-to-VM communication, both Kali and Ubuntu needed to be on a reachable network.

5.4 Live Observation of SSH Authentication Logs

After network troubleshooting, SSH activity was generated and monitored directly on Ubuntu using the authentication log.

```
sudo tail -f /var/log/auth.log
```

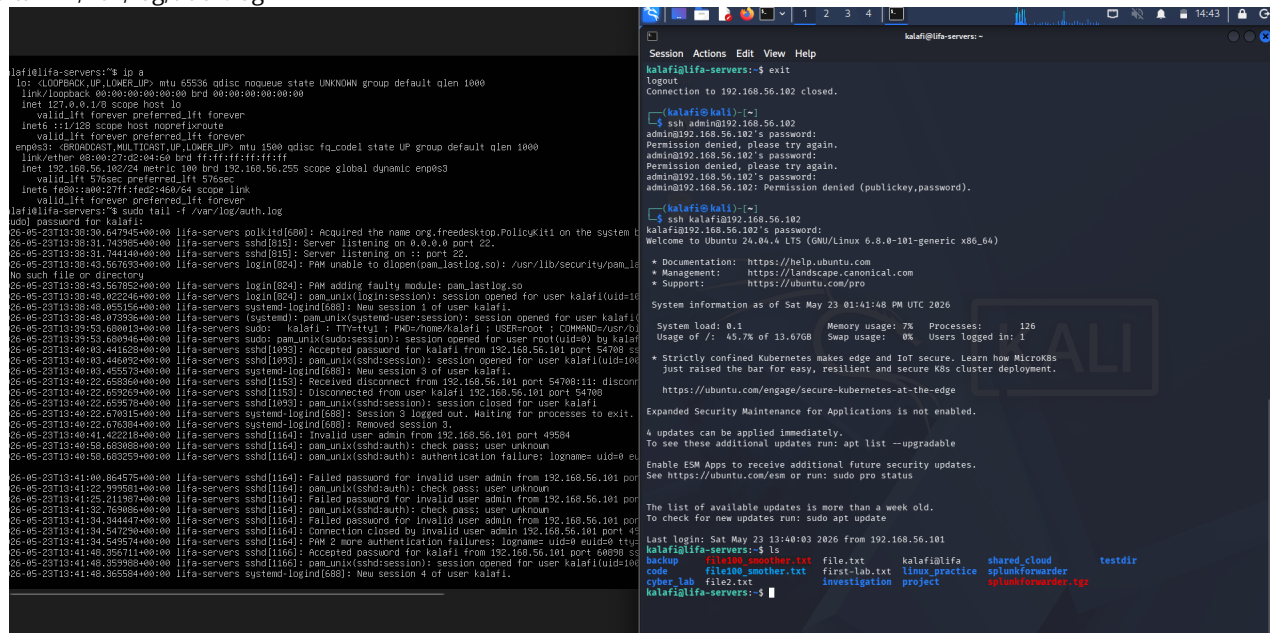


Figure 5: Live SSH authentication logging using `tail -f /var/log/auth.log`.

The live log showed invalid user attempts, failed password attempts, and a successful login for user kalafi. This confirmed that SSH authentication activity was being written locally to `/var/log/auth.log`.

5.5 Splunk Forwarder Status

Although local logs were being generated, new events were not appearing in Splunk as expected. The Splunk Universal Forwarder status was checked.

```
sudo /home/kalafi/splunkforwarder/bin/splunk status
```

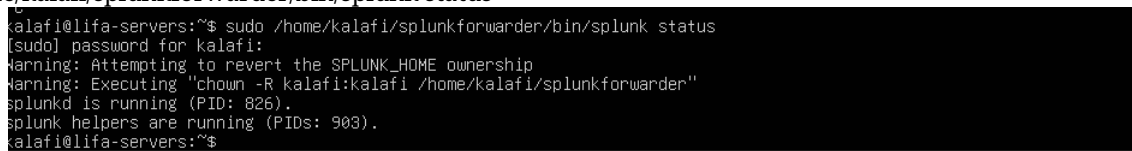


Figure 6: Splunk Universal Forwarder status showing that `splunkd` and helper processes are running.

The forwarder service was running, which meant the problem was not caused by the forwarder being stopped.

5.6 Forward-Server Configuration Check

The next step was to check whether the forwarder was actively connected to Splunk Enterprise.

```
sudo /home/kalafi/splunkforwarder/bin/splunk list forward-server
```

```

kalafi@lifa-servers:~$ sudo /home/kalafi/splunkforwarder/bin/splunk status
[sudo] password for kalafi:
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R kalafi:kalafi /home/kalafi/splunkforwarder"
splunkd is running (PID: 826).
splunk helpers are running (PIDs: 903).
kalafi@lifa-servers:~$ sudo /home/kalafi/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R kalafi:kalafi /home/kalafi/splunkforwarder"
Your session is invalid. Please login.
Splunk username: kalafi
Password:
Active forwards:
    None
Configured but inactive forwards:
    10.225.156.46:9997
kalafi@lifa-servers:~$ _

```

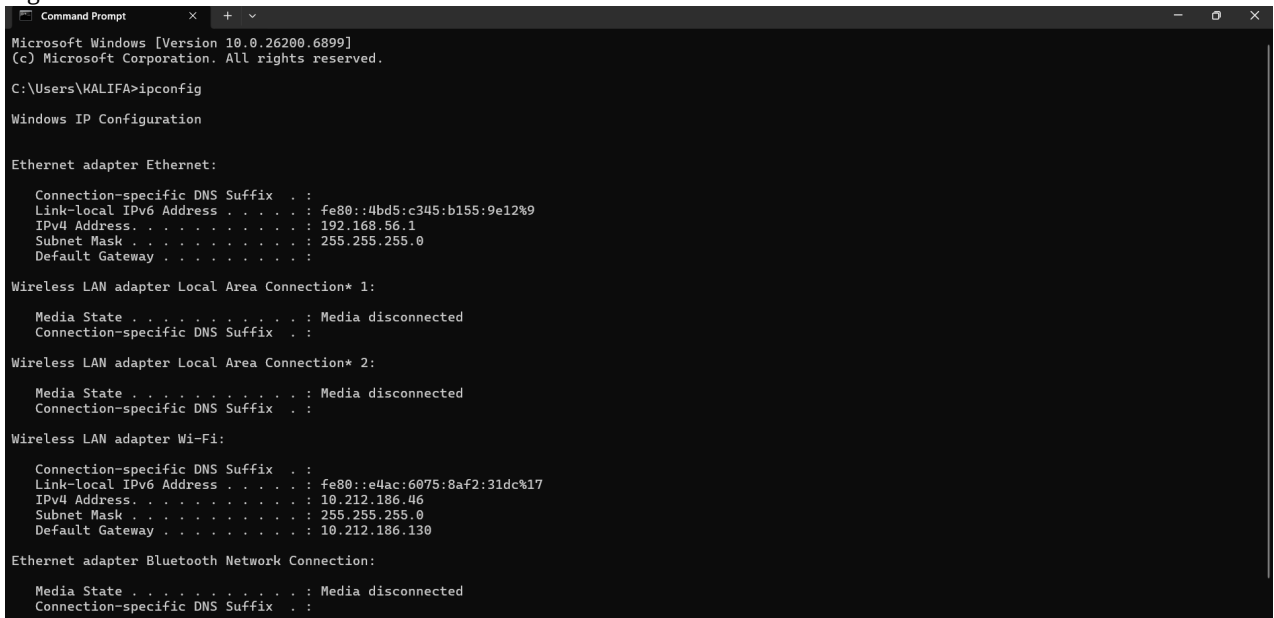
Figure 7: Splunk forward-server status showing no active forwards and one configured but inactive forward.

The output showed no active forwards and one configured but inactive forward to 10.225.156.4:9997. This explained why new logs were not reaching Splunk.

5.7 Identifying the Correct Windows Host-only IP Address

The Windows host IP address on the Host-only network was identified using ipconfig.

ipconfig



```

Microsoft Windows [Version 10.0.26200.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KALIFA>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::4bd5:c345:b155:9e12%9
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::e4ac:6075:8af2:31dc%17
    IPv4 Address. . . . . : 10.212.186.46
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.212.186.130

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

```

Figure 8: Windows ipconfig output showing the Host-only network IP address as 192.168.56.1.

The Windows host IP on the Host-only network was 192.168.56.1. This was the correct address for the Ubuntu forwarder to use when sending logs to Splunk Enterprise.

5.8 Updating the Splunk Forward-Server

The old inactive forward-server was removed, the correct receiver IP was added, and the forwarder was restarted.

```

sudo /home/kalafi/splunkforwarder/bin/splunk remove forward-server 10.225.156.4:9997
sudo /home/kalafi/splunkforwarder/bin/splunk add forward-server 192.168.56.1:9997
sudo /home/kalafi/splunkforwarder/bin/splunk restart
sudo /home/kalafi/splunkforwarder/bin/splunk list forward-server

```

```

kalafi@lifa-servers:~$ sudo /home/kalafi/splunkforwarder/bin/splunk remove forward-server 10.225.156.46:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R kalafi:kalafi /home/kalafi/splunkforwarder"
Stopped forwarding to: 10.225.156.46:9997.
kalafi@lifa-servers:~$ sudo /home/kalafi/splunkforwarder/bin/splunk add forward-server 192.168.56.1:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R kalafi:kalafi /home/kalafi/splunkforwarder"
Added forwarding to: 192.168.56.1:9997.
kalafi@lifa-servers:~$ sudo /home/kalafi/splunkforwarder/bin/splunk restart
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R kalafi:kalafi /home/kalafi/splunkforwarder"
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.

Stopping splunk helpers...

Done.
splunkd.pid doesn't exist...

splunk> Another one.

Checking prerequisites...
Checking mgmt port [8089]: open
New certs have been generated in '/home/kalafi/splunkforwarder/etc/auth'.
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from '/home/kalafi/splunkforwarder/splunkforwarder-10.2.2-80b90d638de6-linux-amd64-manifest'
All installed files intact.
Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

kalafi@lifa-servers:~$ sudo /home/kalafi/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R kalafi:kalafi /home/kalafi/splunkforwarder"
Your session is invalid. Please login.
Splunk username: kalafi
Password:
Active forwards:
192.168.56.1:9997
Configured but inactive forwards:
None
kalafi@lifa-servers:~$

```

Figure 9: Splunk forward-server configuration updated to the active receiver IP address.

After the update, the forward-server list showed an active forward to 192.168.56.1:9997. This confirmed that the forwarder had reconnected to Splunk Enterprise.

5.9 Testing Connectivity to Splunk Receiving Port 9997

Network connectivity from Ubuntu to the Splunk receiver was tested using nc.

```

nc -vz 192.168.56.1 9997
kalafi@lifa-servers:~$ nc -vz 192.168.56.1 9997
Connection to 192.168.56.1 9997 port [tcp/*] succeeded!
kalafi@lifa-servers:~$

```

Figure 10: Network connectivity test showing successful connection from Ubuntu to Splunk Enterprise on port 9997.

The connection succeeded, confirming that Ubuntu could reach the Splunk receiver on port 9997.

5.10 Confirming New Logs in Splunk

After updating the forward-server, new SSH activity was generated and Splunk was searched using a recent time range.

```

index=* host=lifa-servers source="/var/log/auth.log"
| stats count by host source sourcetype

```

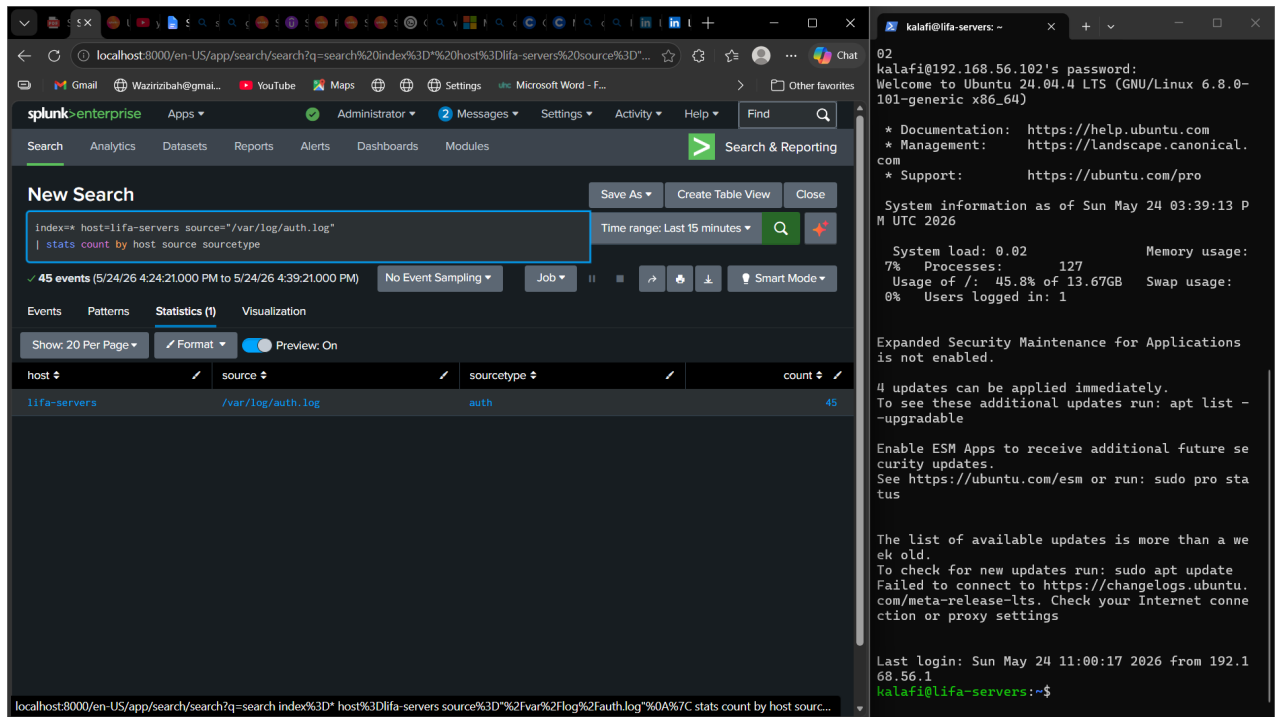


Figure 11: New SSH authentication logs received in Splunk under source `/var/log/auth.log` and sourcetype `auth`.

The Splunk result showed recent events under sourcetype `auth`, confirming that new authentication logs were being forwarded correctly after the fix.

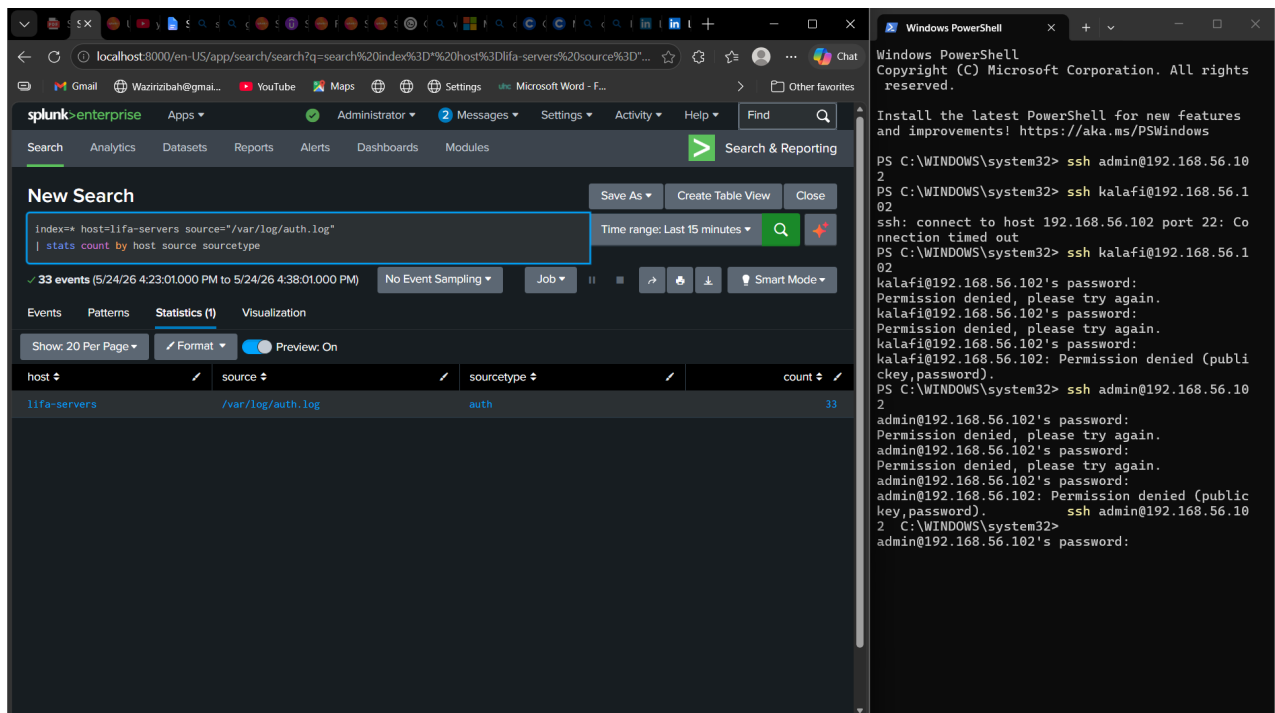


Figure 12: SSH login testing and Splunk count evidence after forwarding was restored.

The screenshot shows SSH login activity being generated from the client side while Splunk recorded new recent events under the `auth` sourcetype.

6. Key Findings

- The SSH service was active and listening on port 22 on the Ubuntu server.
- The first SSH connection attempt from Kali failed with "Network is unreachable," indicating a network connectivity or routing issue.
- After reviewing the VM network adapter settings, SSH activity could be generated and observed locally.
- The Ubuntu authentication log `/var/log/auth.log` captured failed login attempts, invalid user attempts, and successful SSH logins.
- Splunk Universal Forwarder was running, but the forward-server was initially configured but inactive.
- The inactive forward-server was pointing to an old IP address: 10.225.156.4:9997.
- The correct Windows Host-only IP address was identified as 192.168.56.1.
- After updating the forward-server to 192.168.56.1:9997, the forwarder became active.
- The nc test confirmed successful connectivity to Splunk Enterprise on port 9997.
- New SSH authentication logs were received in Splunk under the correct sourcetype auth.

7. SOC Analyst Interpretation

From a SOC analyst perspective, this project demonstrates the importance of separating different layers of a problem. The first layer was the SSH service itself. Because SSH was active and listening, the investigation moved to the network layer. The "Network is unreachable" error indicated that the client could not reach the server network. After SSH activity was restored, the next layer was log forwarding. Although Linux logs were being generated locally, Splunk was not receiving new logs because the Universal Forwarder was configured with an old inactive receiver IP address.

The observed authentication logs also showed security-relevant events such as invalid user attempts, failed passwords, and successful logins. In a real SOC environment, invalid user attempts and repeated failures should be monitored because they may indicate username guessing, brute-force attempts, or unauthorized access attempts.

8. Limitations

- This was a local lab project with a small number of test events.
- The IP addresses were private lab IP addresses and may change when VM network settings change.
- The screenshots provide evidence of troubleshooting steps, but the lab does not represent a full production SOC environment.
- Only SSH authentication and Splunk forwarding were examined in this project.

9. Future Improvements

- Create a dedicated Splunk index for SSH troubleshooting logs.
- Build a Splunk dashboard showing SSH failures, successes, invalid users, and source IPs.
- Create alerts for repeated failed SSH attempts or successful login after multiple failures.
- Document a standard troubleshooting checklist for service, network, log generation, and forwarding issues.
- Extend the project to monitor sudo activity and privilege escalation after successful SSH login.

10. Conclusion

This project showed how to troubleshoot SSH connectivity and Splunk log forwarding issues using a structured approach. The SSH service was confirmed to be running, but the client initially could not reach the Ubuntu server due to a network connectivity issue. After local SSH logs were generated successfully, Splunk did not

receive new events because the Universal Forwarder was still pointing to an old inactive receiver IP. The issue was resolved by identifying the correct Windows Host-only IP address, updating the forward-server configuration to 192.168.56.1:9997, restarting the forwarder, and confirming connectivity with nc. New SSH authentication logs were then received in Splunk under sourcetype auth.

Through this project, key beginner SOC skills were practiced, including Linux service checking, network troubleshooting, live log monitoring, Splunk Universal Forwarder investigation, and validation of log ingestion in Splunk.

Appendix A: Main Commands Used

```
# Check SSH service status
```

```
sudo systemctl status ssh
```

```
# Check Ubuntu IP address
```

```
ip a
```

```
# Monitor live SSH authentication logs
```

```
sudo tail -f /var/log/auth.log
```

```
# Check Splunk Universal Forwarder status
```

```
sudo /home/kalafi/splunkforwarder/bin/splunk status
```

```
# Check forward-server status
```

```
sudo /home/kalafi/splunkforwarder/bin/splunk list forward-server
```

```
# Remove old forward-server
```

```
sudo /home/kalafi/splunkforwarder/bin/splunk remove forward-server 10.225.156.4:9997
```

```
# Add correct forward-server
```

```
sudo /home/kalafi/splunkforwarder/bin/splunk add forward-server 192.168.56.1:9997
```

```
# Restart forwarder
```

```
sudo /home/kalafi/splunkforwarder/bin/splunk restart
```

```
# Test Splunk receiving port
```

```
nc -vz 192.168.56.1 9997
```

Appendix B: Main SPL Queries Used

```
#####
```

```
# Project: Troubleshooting SSH Connectivity and Splunk Forwarding Issues
```

```
# Purpose: SPL queries used to confirm SSH logs and forwarding restoration
```

```
# Data Source: /var/log/auth.log
```

```
# Index: any index searched with index=*
```

```
# Main sourcetype after fix: auth
```

```
#####
```

```
#####
```

```
# Query 1: Confirm log sources, hosts, and sourcetypes
```

```
#####
```

```
index=*
```

```
| stats count by host source sourcetype
```

```
| sort - count
```

```
#####  
# Query 2: Confirm recent auth.log events from the Ubuntu server  
#####  
index=* host=lifa-servers source="/var/log/auth.log"  
| stats count by host source sourcetype
```

```
#####  
# Query 3: Confirm recent SSH logs in Splunk under sourcetype auth  
#####  
index=* host=lifa-servers source="/var/log/auth.log" sourcetype=auth sshd  
| table _time host source sourcetype _raw  
| sort - _time
```

```
#####  
# Query 4: Search failed and successful SSH events after forwarding was restored  
#####  
index=* host=lifa-servers source="/var/log/auth.log" sourcetype=auth ("Failed password" OR "Accepted password" OR  
"Invalid user")  
| rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+\.\d+)"  
| table _time host src_ip sourcetype _raw  
| sort - _time
```

```
#####  
# Query 5: Summarize failed and successful SSH activity by source IP  
#####  
index=* host=lifa-servers source="/var/log/auth.log" sourcetype=auth ("Failed password" OR "Accepted password" OR  
"Invalid user")  
| rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+\.\d+)"  
| stats count(eval(searchmatch("Failed password"))) as failed_password_events,  
count(eval(searchmatch("Invalid user"))) as invalid_user_events,  
count(eval(searchmatch("Accepted password"))) as successful_login_events  
by src_ip  
| sort - failed_password_events
```